

Software Engineering (UE24CS341A)

Lab 1: Requirements Engineering & UML Use-Case Modelling

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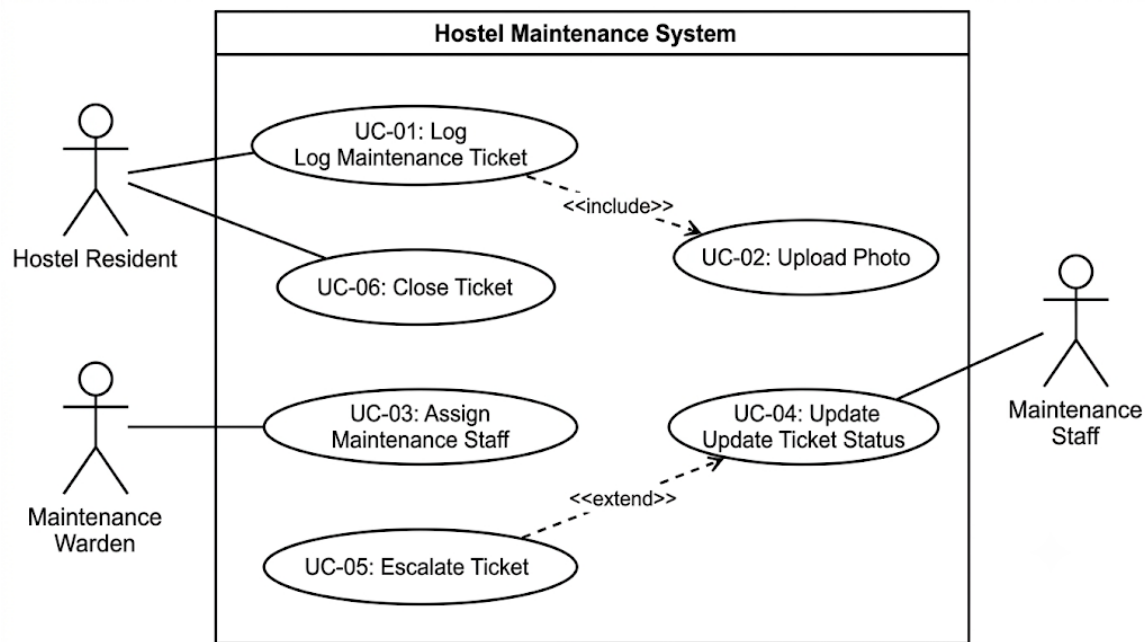
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Lab 1: Hostel Maintenance & Issue Ticketing System

1. Requirements Table

Req ID	Type	Description	Priorit	Acceptance	Rationale
FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: Ticket is created, assigned a unique ID, and routed to warden queue; Fail: Incomplete form accepted without room tag.	Core functionality for issue reporting
FR-002	Functional	The system shall allow the Maintenance Warden to assign an open ticket to a specific Maintenance Staff member.	High	Pass: Warden can select staff from a dropdown and ticket status changes to Assigned.	Necessary for dispatching staff
FR-003	Functional	The system shall allow Maintenance Staff to update the status of an assigned ticket (e.g., In Progress, Resolved).	High	Pass: Staff updates status to Resolved and timestamp is recorded.	Tracks progress of the issue
FR-004	Functional	The system shall require the Hostel Resident to digitally sign off to close a resolved ticket.	Medium	Pass: Ticket status changes to Closed after resident confirms.	Ensures resident satisfaction before closure
FR-005	Functional	The system shall display a dashboard of all active tickets for the Maintenance Warden, sortable by severity and time.	Medium	Pass: Warden sees a list of tickets sortable by severity.	Allows warden to oversee and prioritize work
NFR-001	Non-functional	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures urgent issues are not ignored
NFR-002	Non-functional	The system shall have an uptime of 99.9% during the academic semester.	Medium	Pass: Uptime logs show <= 43 minutes of downtime per month.	Residents must be able to report urgent issues

2. UML Use-Case Diagram



3. Actors & Use Cases

Actors: Hostel Resident, Maintenance Warden, Maintenance Staff

Use-Case Flow Specification

Use Case: UC-02 : Assign Maintenance Staff

Primary Actor: Maintenance Warden

Preconditions: A maintenance ticket has been created and is available in the Maintenance Warden's queue. The Maintenance Staff member is registered in the system.

Postconditions: The maintenance ticket is assigned to a Maintenance Staff member and the assignment is recorded in the system.

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Main Success Scenario:

1. Warden navigates to the "**Maintenance Tickets**" section.
2. System displays the list of pending maintenance tickets.
3. Warden selects a ticket that requires staff assignment.

4. System displays the ticket details, including room number, issue category, severity, description, and photo.
5. Warden selects "**Assign Maintenance Staff**".
6. System displays a list of available Maintenance Staff members.
7. Warden selects an appropriate Maintenance Staff member based on the issue category and availability.
8. Warden confirms the staff assignment.
9. System assigns the selected Maintenance Staff member to the ticket and updates the ticket status.
10. System displays a confirmation message indicating that the ticket has been successfully assigned.

Alternate Flow:

6a. No Maintenance Staff Available

1. System displays a message indicating that no Maintenance Staff members are currently available.
2. Warden is prompted to either wait for staff availability or keep the ticket in the pending queue.
3. System retains the ticket as "**Unassigned**".
4. Use case ends until a Maintenance Staff member becomes available.